

## MAIN-BELT ASTEROID

## 4 Vesta

**AVERAGE DISTANCE TO SUN** 219 million miles (353 million km)

**ORBITAL PERIOD** 3.63 years

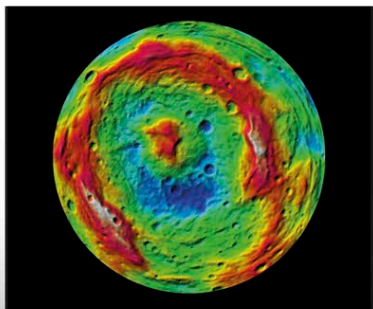
**ROTATION PERIOD** 5.34 hours

**DIAMETER** 330 miles (530 km)

**DATE OF DISCOVERY** March 29, 1807

Vesta is one of the largest asteroids, with a surface that reflects, on average, 42 percent of the incoming light. This makes it the brightest asteroid in the night sky and the only one that is visible to the unaided eye. Most asteroids of Vesta's size are expected to be nearly spherical, but Vesta's shape has been distorted by a massive impact at the south pole that created a huge basin called Rheasilvia, which measures some 300 miles (500 km) across—almost as wide as Vesta itself. Like many

large craters on the Moon, Rheasilvia has a mountainous central peak. It overlaps an older and slightly smaller crater, which is 230 miles (375 km) wide. Some of the fragments of Vesta's crust produced by the cratering process are still trailing Vesta on similar orbits. Others have hit Earth and been recognized as strange meteorites with a composition similar to the igneous rock basalt. Six percent of Earth's recent meteorite falls are Vesta-like in



their mineralogical makeup. The composition is similar to the lava that spews out of Hawaiian volcanoes. In its early life, Vesta melted and then resolidified, with denser material sinking to the center. It now has a layered structure like the rocky planets, with a low-density crust lying above layers of pyroxene and olivine and an iron core. Vesta is thought to be the only remaining differentiated asteroid in the Main Belt. It is one of the densest asteroids known, with a density similar to that of Mars. In July 2011, NASA's Dawn spacecraft went into orbit around Vesta, beginning a year-long study of the asteroid. Vesta's

## SOUTH-POLAR IMPACT BASIN

A giant impact basin called Rheasilvia at the south pole of Vesta is seen here in a false-color image taken by NASA's Dawn spacecraft. It is 300 miles (500 km) across and has a central peak (shown in red) and a rim that is 9 miles (15 km) high.



## METEORITE

This meteorite, which landed in Western Australia in October 1960, originated from Vesta.

surface turned out to be old and heavily cratered, with grooves running around the equator, possibly fractures from the south-polar impact. One of the asteroid's most distinctive features was a chain of three craters nicknamed the Snowman, seen in the image below. From top to bottom, the individual craters are called Minucia, which is 13 miles (21.5 km) wide; Calpurnia, 30 miles (50 km) wide; and Marcia, 36 miles (58 km) wide. The Dawn spacecraft's instruments are studying the surface composition of Vesta, with the objective of matching meteorites to specific areas on the asteroid. One likely source is the south-polar mountain, which rises 13 miles (22 km) within the Rheasilvia impact basin.

## SNOWMAN CRATERS

The three craters that form the "Snowman" are at the bottom left in this image. The smoother surface around them is thought to be a blanket of debris thrown out by the impacts.

